

## # SKYCREW – Roadmap, Open-Source Dependencies &amp; Risk Register

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## ## 1. Phased build plan

| Phase                   | Goal                                                                           | Duration | Exit criteria                                                                        |
|-------------------------|--------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------|
| **0 – Design freeze**   | This document set; component selection; venue-partner pilot interest           | 1–2 mo   | BOM locked; 5+ pilot partners signed LOI                                             |
| **1 – Prototype**       | 1 Sparrow + 4 UWB anchors + OpenHD link + orchestration MVP                    | 3–4 mo   | **Gate: indoor hover/hold <0.10 m**, dispatch a pod trigger → drone → angle in <15 s |
| **2 – Pilot**           | 3–5 Certified Partners run real events; telemetry + bug portal live            | 3–4 mo   | ≥15 events, 0 serious incidents, mean-time-between-fault documented                  |
| **3 – Crowdfund**       | Kickstarter (34 days, \$50K goal) → Indiegogo InDemand                         | 2 mo     | \$250K+ raised; 1,700+ backers                                                       |
| **4 – First batch**     | COTS build; fulfillment waves (Sparrow → Condor → Duo)                         | 6–8 mo   | post-campaign   All kits shipped, NPS > 50                                           |
| **5 – Custom hardware** | SKYCREW carbon airframe + custom FC carrier board (funded by batch 1 margin)   | 9–14 mo  | Airframe cert, weight -15%, cost -10%                                                |
| **6 – Webapp GA**       | Venue Planner + Edit module + live stream + Fund-a-Guest menu (stretch-funded) | 9–18 mo  | 100 active partners; 30% of events use streaming; 25% offer Fund-a-Guest tiers       |
| **7 – Hydrogen v2**     | Fuel-cell hybrid prototype (Harris Aerial / IE-SOAR 2.4 kW class)              | 18+ mo   | 2.5 h single-aircraft endurance, NDAA-compliant                                      |

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## ## 2. Open-source / open-hardware dependency list

| Component                | Project                                                                           | License                                 | Use                                                                    |
|--------------------------|-----------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------|
| Flight stack             | ArduPilot / ArduCopter (Skybrush CMcopter fork)                                   | GPL-3.0                                 | Autonomous flight, failsafes, MAVLink                                  |
| Drone-show orchestration | Skybrush Live + Studio + Server                                                   | GPL/OSS                                 | Fleet choreography, Blender shot design, trajectory validation         |
| Video + telemetry link   | OpenHD                                                                            | GPL-3.0                                 | Sub-100 ms video, MAVLink, RC-over-link                                |
| Gimbal                   | STorm32 BGC (open HW) · DawnBGC (MIT) · SimpleFOC                                 | TAPR-OHL / MIT                          | 2- and 3-axis stabilization                                            |
| Indoor positioning       | DWM1000/Qorvo UWB + PMW3901 optical flow; ArduPilot `GPS_INPUT` ext-position path | HW + GPL                                | Indoor absolute 3D position (Vicon-equivalent workflow, cheap anchors) |
| Trigger pods             | ESP32 firmware (ours, OSS)   MIT (proposed)                                       | VAD + button focus requests; sync audio |                                                                        |
| Ground control           | QGroundControl / Mission Planner                                                  | GPL/Apache                              | Telemetry, preflight, geofence                                         |
| Streaming                | OBS + LiveKit/RTMP                                                                | GPL/Apache                              | Operator laptop stream-out + remote-guest video wall                   |
| Webapp                   | Three.js (Skybrush Viewer pattern), COLMAP/OpenSfM photogrammetry                 | MIT/GPL                                 | Venue planner rendering + 3D model building                            |

> Licensing note: GPL-3.0 components (ArduPilot, OpenHD) impose copyleft on \*derived\* firmware/software. SKYCREW's proprietary value is the \*\*orchestration layer, pod-dispatch logic, and webapp\*\* – cleanly separated as an application layer talking to the GPL stack over MAVLink/REST, preserving the option to license the orchestrator commercially.

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## ## 3. Risk register (summary)

| #  | Risk                          | L × I       | Mitigation                              | Owner |
|----|-------------------------------|-------------|-----------------------------------------|-------|
| R1 | Indoor positioning gate fails | High × High | Phase-1 kill criteria; UWB+optical-flow |       |

fusion is published+reproducible; fallback to outdoor-first v1 | Eng |  
 | R2 | Part 107 / crowd-flight rules block events | Med × High | Enclosed props, geofence, no-over-crowd SOP, §107.35 waiver pack, ≤250g option; indoor venue ops may sit outside FAA airspace but liability still managed | Ops/Legal |  
 | R3 | Rotor noise ruins toasts | Med × Med | Quiet cinewhoop props, height discipline, shot-window scheduling | Eng |  
 | R4 | Venue Wi-Fi interference | Med × Med | OpenHD broadcast mode + frequency plan per venue; dual-band; site survey step | Ops |  
 | R5 | Crowdfund under-delivery | Med × High | \$50K goal (high-success band), COTS no-tooling first batch, conservative 6–8 mo ship | PM |  
 | R6 | Battery logistics overwhelm single operator | Low × Med | Auto-RTL failsafe, pad-per-drone, 8-pack rotation, pad landing is fully automated | Ops |  
 | R7 | FX3/BMPCC are expensive single point | Low × Med | Customer-owned-payload option; POD A fallback; hull insurance | PM |  
 | R8 | Data privacy / footage misuse | Med × Med | Venue consent contract, signage, operator holds footage, deletion clauses | Legal |  
 | R9 | GPL compliance drift | Low × Med | Legal review of separation strategy (see licensing note) | Legal |

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#### ## 4. What success looks like (18 months)

- 100+ active partners (rent / lease / franchise) across US + EU.
- 500+ events filmed; zero serious incidents.
- 30% of events use the live-stream/video-wall add-on (recurring revenue).
- 25% of events sell Fund-a-Guest tiers (avg 15+ items/event) – a self-funding-film story that drives word-of-mouth.
- Crowdfunding delivered on time with NPS > 50.
- Custom airframe shipping; hydrogen v2 prototype flying.
- Community portal with 1,000+ members, public roadmap, beta ring active.